

NCERT Class 10 Science - Chapter 9: Light - Reflection and Refraction

9.1 Reflection of Light & Laws of Reflection

Laws of Reflection: (i) The angle of incidence is equal to the angle of reflection (angle i = angle r). (ii) The incident ray, normal at the point of incidence and reflected ray lie in the same plane.

9.2.2 Sign Convention for Spherical Mirrors

New Cartesian Sign Convention: Pole P is origin. Object distance u is always negative. Focal length f of a concave mirror is ALWAYS negative ($-f$). Focal length f of a convex mirror is ALWAYS positive ($+f$).

9.2.3 Mirror Formula and Magnification

Mirror Formula: $1/f = 1/v + 1/u$. Magnification $m = h'/h = -v/u$. Negative m implies real inverted image. Positive m implies virtual erect image.

9.3 Refraction of Light & Snell's Law

Snell's Law: The ratio of sine of angle of incidence to sine of angle of refraction is constant: $\sin(i) / \sin(r) = n_{21}$ (refractive index). Light traveling from rarer to denser medium bends towards the normal.

9.3.5 Lens Formula and Magnification

Spherical Lens Formula: $1/f = 1/v - 1/u$ (Note minus sign!). Lens Magnification $m = h'/h = +v/u$ (Positive sign, unlike mirror formula). Focal length of convex lens is positive, concave lens is negative. Power of lens $P = 1/f(\text{metres})$ in Dioptres (D). For a convex lens placed between an object and a screen, a real inverted image is formed on the screen when $u > f$.